

A Technical Introduction to Regularization machine learning

Section 1

$$w_T = \gamma n \sum_{i=0}^{T-1} (I - \gamma n X^T X)^i X^T Y^{\wedge} \quad \{displaystyle w_T = \frac{\gamma n}{\sum_{i=0}^{T-1} (I - \frac{\gamma n}{\gamma n} \hat{X})^i} \mathsf{T} \hat{X} \hat{Y}\}$$

Section 2

$R(f) = f^T L f$ $\{displaystyle R(f) = \bar{f}^T \mathsf{L} \bar{f}\}$ where $L = D - W$ $\{displaystyle L = D - W\}$ is the Laplacian matrix of the graph induced by W $\{displaystyle W\}$. The optimization problem $\min_{f \in \mathbb{R}^m} R(f), m = u + l$ $\{displaystyle \min_{f \in \mathbb{R}^m} R(f), m = u + l\}$ can be solved analytically if the constraint $f(x_i) = y_i$ $\{displaystyle f(x_i) = y_i\}$ is applied for all supervised samples. The labeled part of the vector f $\{displaystyle f\}$ is therefore obvious. The unlabeled part of f $\{displaystyle f\}$ is solved for by:

Section 3

$$R(w) = \sum_{g=1}^G \|w_g\|_2, \quad \{displaystyle R(w) = \sum_{g=1}^G \|w_g\|_2\} \text{ where } \|w_g\|_2 = \sum_{j=1}^{|G_g|} (w_{gj})^2 \quad \{displaystyle \|w_g\|_2 = \sqrt{\sum_{j=1}^{|G_g|} (w_{gj})^2}\}$$

Section 4

$$\min_{w \in \mathbb{R}^{p+1}} \|Xw - Y\|_2^2 + \lambda (\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2), \quad \alpha \in [0, 1] \quad \{displaystyle \min_{w \in \mathbb{R}^{p+1}} \frac{1}{2} \|Xw - Y\|_2^2 + \lambda (\alpha \|w\|_1 + (1 - \alpha) \|w\|_2^2), \alpha \in [0, 1]\}$$

Section 5

$$w_T = (I - \gamma n X^T X)^{\gamma n \sum_{i=0}^{T-2} (I - \gamma n X^T X)^i} X^T Y^{\wedge} + \gamma n \sum_{i=1}^{T-1} (I - \gamma n X^T X)^{i-1} X^T Y^{\wedge} + \gamma n \sum_{i=0}^{T-1} (I - \gamma n X^T X)^i X^T Y^{\wedge} \quad \{displaystyle \begin{aligned} w_T &= \left(I - \frac{\gamma n}{\gamma n} \hat{X} \right)^{\frac{\gamma n}{\gamma n} \sum_{i=0}^{T-2} \left(I - \frac{\gamma n}{\gamma n} \hat{X} \right)^i} \mathsf{T} \hat{X} \hat{Y} + \frac{\gamma n}{\gamma n} \hat{X} \hat{Y} + \frac{\gamma n}{\gamma n} \sum_{i=1}^{T-1} \left(I - \frac{\gamma n}{\gamma n} \hat{X} \right)^{i-1} \mathsf{T} \hat{X} \hat{Y} \\ &= \frac{\gamma n}{\gamma n} \sum_{i=0}^{T-1} \left(I - \frac{\gamma n}{\gamma n} \hat{X} \right)^i \mathsf{T} \hat{X} \hat{Y} \end{aligned}$$

Section 6

Regularizers for sparsity Assume that a dictionary ϕ_j $\{displaystyle \phi_j\}$ with dimension p $\{displaystyle p\}$ is given such that a function in the function space can be expressed as:

Section 7

Enforcing a sparsity constraint on w $\{displaystyle w\}$ can lead to simpler and more interpretable models. This is useful in many real-life applications such as computational biology. An example is developing a simple predictive test for a disease in order to minimize the cost of performing medical tests while maximizing predictive power. A sensible sparsity constraint is the L_0 $\{displaystyle L_0\}$ norm $\|w\|_0$ $\{displaystyle \|w\|_0\}$, defined as the number of non-zero elements in w $\{displaystyle w\}$. Solving a L_0 $\{displaystyle L_0\}$ regularized learning problem, however, has been demonstrated to be NP-hard. The L_1 $\{displaystyle L_1\}$ norm (see also Norms) can be used to approximate the optimal L_0 $\{displaystyle L_0\}$ norm via convex relaxation. It can be shown that the L_1 $\{displaystyle L_1\}$ norm induces sparsity. In the case of least squares, this problem is known as LASSO in statistics and basis pursuit in signal processing.

Section 8

L_1 $\{displaystyle L_1\}$ regularization can occasionally produce non-unique solutions. A simple example is provided in the figure when the space of possible solutions lies on a 45 degree line. This can be problematic for certain applications, and is overcome by combining L_1 $\{displaystyle L_1\}$ with L_2 $\{displaystyle L_2\}$ regularization in elastic net regularization, which takes the following form:

Section 9

Regularization in machine learning In machine learning, a key challenge is enabling models to accurately predict outcomes on unseen data, not just on familiar training data. Regularization is crucial for addressing overfitting—where a model memorizes training data details but cannot generalize to new data. The goal of regularization is to encourage models to learn the broader patterns within the data rather than memorizing it. Techniques like early stopping, L1 and L2 regularization, and dropout are designed to prevent overfitting and underfitting, thereby enhancing the model's ability to adapt to and perform well with new data, thus improving model generalization.

Section 10

$$R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2 : w = \sum_{g=1}^G w_g \right\} \quad \{displaystyle R(w) = \inf \left\{ \sum_{g=1}^G \|w_g\|_2 : w = \sum_{g=1}^G \bar{w}_g \right\}$$

Section 11

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

Section 12

where X and Y are the domains of input data x and their labels y respectively. Typically in learning problems, only a subset of input data and labels are available, measured with some noise. Therefore, the expected error is unmeasurable, and the best surrogate available is the empirical error over the N available samples:

Section 13

Classification Empirical learning of classifiers (from a finite data set) is always an underdetermined problem, because it attempts to infer a function of any x given only examples $x_1, x_2, \dots, x_n$. A regularization term (or regularizer) $R(f)$ is added to a loss function:

Section 14

If W_{ij} encodes the result of some distance metric for points x_i and x_j , it is desirable that $f(x_i) \approx f(x_j)$. This regularizer captures this intuition, and is equivalent to: